

MOBILE BROOKLEY, AL, USA

WMO: 722235

Lat: 30.626N

Lon: 88.068W

Elev: 8

StdP: 101.23

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13838

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.6	0.6	-9.9	1.6	2.6	-7.4	2.0	4.5	10.7	6.3	9.3	8.6	3.4	330	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	34.3	25.8	33.3	25.6	32.5	25.4	27.5	31.5	27.0	30.9	26.6	30.5	3.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.3	26.1	21.5	29.1	25.5	20.8	28.8	87.0	31.7	84.7	31.0	82.6	30.3	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	8.0	7.2	-4.9	36.1	2.0	1.4	-6.3	37.1	-7.5	38.0	-8.6	38.8	-10.1	39.8		
			-6.3	28.4	1.8	0.7	-7.6	28.9	-8.6	29.3	-9.6	29.7	-10.9	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	10.7	12.7	16.1	19.7	23.8	27.0	28.0	28.1	26.3	21.2	15.6	12.2
	DBStd	7.23	5.18	4.65	4.24	3.24	2.70	1.70	1.41	1.56	2.32	4.03	4.34	5.00
	HDD10.0	126	56	23	6	0	0	0	0	0	0	0	7	35
	HDD18.3	843	241	163	93	24	2	0	0	0	0	19	104	198
	CDD10.0	3829	76	100	194	290	427	511	558	561	489	348	173	102
	CDD18.3	1504	3	6	23	63	170	261	300	302	239	110	21	7
	CDH23.3	14213	2	12	76	308	1397	2692	3248	3278	2292	818	81	9
(13)	CDH26.7	5160	0	0	6	27	370	1045	1344	1365	809	188	6	0
(14)	Wind	WSAvg	3.2	3.5	3.5	3.6	3.6	3.2	2.7	2.4	2.5	3.1	3.3	3.4
(15) Precipitation	PrecAvg	1672	139	114	156	138	126	153	209	166	137	85	115	132
	PrecMax	2152	408	260	400	449	378	452	577	325	552	241	297	368
	PrecMin	1072	25	19	16	16	9	42	98	28	50	1	34	32
	PrecStd	295	91	62	87	105	82	89	93	77	107	65	73	70
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	25.1	27.4	28.8	33.0	35.1	35.7	35.8	34.9	31.7	27.5	24.6
		MCWB	19.4	19.8	19.2	21.4	23.1	25.0	26.2	25.7	25.5	24.1	22.7	20.6
	2%	DB	21.4	22.7	25.5	27.4	31.2	33.5	34.0	34.1	32.8	30.0	25.8	22.8
		MCWB	18.6	19.3	19.5	21.0	23.2	25.2	26.1	26.1	24.8	23.4	21.6	20.1
	5%	DB	20.0	21.2	23.7	26.3	29.8	32.5	33.0	33.0	31.9	28.7	23.9	21.5
		MCWB	17.9	18.4	19.0	20.7	23.2	25.1	25.7	26.0	24.6	22.6	20.1	19.4
	10%	DB	18.6	19.7	22.4	25.2	28.7	31.4	32.1	32.3	30.9	27.4	22.6	19.9
		MCWB	16.9	17.4	18.8	20.4	23.1	24.8	25.7	25.8	24.4	22.3	19.0	18.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.8	21.8	22.4	24.5	26.2	27.6	28.0	28.4	27.7	26.5	24.5	22.3
		MCDB	22.0	23.1	24.6	26.5	29.0	31.8	32.4	32.7	31.0	28.7	26.2	23.4
	2%	WB	19.5	20.5	21.4	23.5	25.3	26.9	27.4	27.6	26.8	25.5	22.7	21.1
		MCDB	20.7	21.9	23.4	25.8	28.7	30.9	31.5	31.7	30.1	28.2	24.6	22.2
	5%	WB	18.5	19.2	20.6	22.6	24.7	26.4	26.8	27.0	26.1	24.6	21.3	19.7
		MCDB	19.5	20.4	22.6	24.9	28.1	30.2	31.0	31.0	29.5	27.4	23.1	21.1
	10%	WB	17.3	18.0	19.6	21.8	24.1	25.8	26.4	26.6	25.6	23.7	20.2	18.5
		MCDB	18.2	19.1	21.5	24.0	27.4	29.6	30.5	30.5	29.0	26.4	21.9	19.6
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	9.8	10.2	9.7	9.3	8.5	8.3	8.4	8.8	10.4	10.9	9.7
		MCDBR	9.8	9.5	10.2	9.9	10.1	9.9	9.4	9.5	9.6	10.1	10.6	9.5
		MCWBR	7.3	6.6	5.9	5.3	4.2	3.8	3.6	3.7	3.8	4.7	6.6	7.4
	5% WB	MCDBR	8.2	8.0	8.0	7.6	7.9	8.3	8.4	8.3	7.8	7.6	8.7	8.7
		MCWBR	7.0	6.6	5.6	5.1	3.9	3.9	3.8	3.8	3.6	4.3	6.7	7.4
(40) Clear-Sky Solar Irradiance	taub		0.331	0.349	0.373	0.403	0.439	0.465	0.514	0.503	0.447	0.383	0.355	0.340
	taud		2.547	2.500	2.427	2.346	2.280	2.258	2.140	2.167	2.291	2.454	2.476	2.531
	Ebn at Noon		902	915	911	891	856	829	787	792	827	865	864	871
	Edh at Noon		85	98	112	126	135	138	155	149	127	101	91	83
(44) All-Sky Solar Radiation	RadAvg		3.04	3.60	4.86	5.93	6.55	6.26	5.91	5.56	5.20	4.54	3.47	2.67
	RadStd		0.24	0.49	0.39	0.47	0.35	0.48	0.40	0.40	0.32	0.47	0.30	0.24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+180
(47) Regional (31 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+93

Nomenclature: See separate page